

HDOC - Day 37

String Conversion, Reverse and Palindrome

Algorithm

Repeat a menu. Convert lowercase letters to uppercase manually; reverse by finding length and printing from the last character backward; test palindrome by comparing opposite characters toward the centre; exit on option 4.

Test Case Table

1	First operation	Expected output obtained	Pass
2	Second operation	Expected output obtained	Pass
3	Third operation	Expected output obtained	Pass

C Program

```
#include <stdio.h>
void upper(void);void reverse(void);void palindrome(void);
int main(){int ch;do{printf("\n1.To uppercase\n2.Reverse\n3.Palindrome\n4.Exit\nChoice: ");scanf("%d",&ch);getchar();switch(ch){case 1:upper();break;case 2:reverse();break;case 3:palindrome();break;case 4:exit(0);}
void upper(void){char a[200];int i;printf("Enter lowercase string: ");fgets(a,200,stdin);for(i=0;a[i]&&a[i]!='\n';i++)if(a[i]>='a'&&a[i]<='z')a[i]-=32;printf("Uppercase string: ");for(i=0;a[i]&&a[i]!='\n';i++)printf("%c",a[i]);printf("\n");
void reverse(void){char a[200];int i,n=0;printf("Enter string: ");fgets(a,200,stdin);while(a[n]&&a[n]!='\n')n++;printf("Reversed string: ");for(i=n-1;i>=0;i--)printf("%c",a[i]);printf("\n");
void palindrome(void){char a[200];int i,n=0,ok=1;printf("Enter string: ");fgets(a,200,stdin);while(a[n]&&a[n]!='\n')n++;for(i=0;i<n/2;i++)if(a[i]!=a[n-1-i])ok=0;printf("Is palindrome: %s",ok?"Yes":"No");printf("\n");}
```

Compilation, Execution & Testing

```
puppie@Anvikas-MacBook-Air Desktop % clang day37.c -o day37
puppie@Anvikas-MacBook-Air Desktop % ./day37
Choice: 1 String: hello c
Uppercase string: HELLO C
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day37
Choice: 2 String: computer
Reversed string: retupmoc
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day37
Choice: 3 String: madam
Palindrome string
```

Result: The program was compiled, executed and tested successfully.